

Context Rot Is a Computation Failure, Not a Storage Failure

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Abstract

A search-free chess transformer loses track of the board as a game lengthens, and its illegal moves are largely admissible under the board it appears to believe. Since that belief is causally read by the policy, writing a better one should fix the behaviour. We test whether it can be written without an oracle, which is the only version of the question that matters outside domains with a rules engine.

It cannot, and the reason is structural rather than incidental. We learn a transition operator on frozen activations, supervised only by the requirement that the model’s own state evolve consistently under the moves it reads. The operator learns its task well, reaching 0.996 accuracy on held-out rollouts. Rolling an earlier state forward nonetheless yields a board that decodes *worse* than the model’s current state at every depth, 0.510 against 0.530 in the opening and 0.346 against 0.362 at depth, and test-time injection changes the illegal-move rate by nothing measurable against no-op, stale and norm-matched random controls. The explanation is that a model at position t already attends to the entire prefix, so its earlier state is a lossy function of the same tokens and reconstruction adds no information. The tokens are not lost. The computation over them is.

Training the consistency objective jointly with the language model is worse than useless: it raises validation loss from 3.0287 to 3.4250 while leaving the consistency task equally learnable.

One positive result survives. The disagreement between the model’s current state and its rolled-forward earlier state is computable at inference with no oracle at any stage, and it carries information the model’s own confidence does not: held out, combining the two reaches AUC 0.6482 against 0.6299 for confidence alone, cutting the illegal-move rate among retained positions from 0.1423 to 0.1283 at half coverage. Detection is available where repair is not.

Keywords: context rot, state tracking, self-supervision, world models, selective prediction

1 Introduction

A companion study of search-free chess transformers established three facts that together make a repair look feasible. Board state is linearly decodable and decays with depth, from 0.975 occupied-square accuracy in the opening to 0.066 at the end of a long game. Illegal moves are largely coherent with the decayed state: 0.355 of them are admissible under the board the model appears to hold, against 0.050 for a mismatched belief. And editing the state representation changes the emitted move, so the policy reads it.

If a wrong action follows a wrong belief, and the belief can be written, then writing a better belief should produce a better action. The obstacle is knowing what to write. An oracle would say, and an oracle is exactly what a deployed system lacks. A method that needs a rules engine is confined to domains that have one, which excludes every setting where context rot actually costs anything.

This paper asks whether the state can be repaired using only what the model already has: its own activations, and the tokens already in its context. The answer is no, and the reason turns out to be more interesting than the attempt.

Contributions.

1. **A negative result with a mechanism.** Oracle-free state repair does not work, because a model at position t already attends to the whole prefix. Its earlier state is a lossy function of those same tokens, so rolling it forward cannot supply information the model does not have. Context rot in this setting is a failure to compute with available information, not a failure to retain it.
2. **Evidence that separates the two ways this could fail.** A learned transition operator, fitted with no oracle, reaches 0.996 on its own task while producing rollouts that decode worse than the state they were meant to repair. The method is refuted rather than merely untested.
3. **A cost of the obvious alternative.** Training the consistency objective jointly with the language model degrades it, 3.0287 to 3.4250 in validation loss, reproducing the pattern by which auxiliary state objectives buy representation quality at the expense of the task.
4. **A signal that does work.** State inconsistency is computable at inference with no oracle at any stage, and adds information beyond the model’s own confidence.

2 Method

The consistency condition. A correct state satisfies a requirement that needs no external supervision: the state after reading token x_{t+1} must be a function of the state before it and of x_{t+1} . We learn a projection $z_t = \text{normalise}(Ph_t)$ into a state subspace and a transition $\hat{z}_{t+1} = \text{normalise}(T[z_t; e(x_{t+1})])$, trained so the model’s own representation satisfies it.

Why the objective is contrastive, and against what. A regression target invites collapse: a constant z satisfies consistency perfectly and carries nothing. We therefore use InfoNCE, and the choice of negatives is what makes the task non-trivial. Two designs failed before the one we use, and both scored highly while representing nothing of interest.

Pooling negatives across the batch lets the model win by encoding *which game* it is in; accuracy passed 0.97 within a thousand steps. Drawing negatives from the same sequence instead lets it win by encoding *position*, since the residual stream carries positional embeddings and the transition need only add one; accuracy reached 0.99. We draw negatives from the *same timestep in other sequences*. Every candidate then shares the position exactly, so a component depending only on position is identical across candidates and cannot discriminate, and game identity is no longer sufficient because the target must be matched at a specific moment.

A third attempt, subtracting the per-position batch mean to remove positional information explicitly, was both unnecessary and wrong: unnecessary because the negatives already neutralise position, and wrong because it leaves the representation undefined for a single sequence. At inference the mean is taken over a batch of one and the projection returns zeros. We record this because it briefly produced an encouraging result that was an artifact of comparing move embeddings against a vanished state.

Frozen rather than joint. We fit the operator on a frozen model. Training it jointly with the language model degrades the language model (Section 3.1), and freezing also makes the method more useful: it applies to a model that already exists, and the repair becomes an inference-time

Table 1: Is a rolled-forward state better than the model’s current one? Board accuracy on occupied squares, decoded from the state subspace at the same positions. The one-step operator compounds error and lands below even the unrolled stale state. Training on rollouts recovers most of that, and still does not exceed the state it was meant to repair.

Ply	one-step operator			multi-step operator		
	current	rolled	stale	current	rolled	stale
20–30	0.499	0.408	0.427	0.530	0.510	0.427
30–40	0.466	0.369	0.425	0.484	0.461	0.444
40–50	0.445	0.336	0.416	0.452	0.436	0.423
50–60	0.407	0.302	0.386	0.416	0.395	0.395
60–80	0.351	0.286	0.334	0.362	0.346	0.340

intervention rather than a training recipe. Only a projection, a move embedding and a linear transition are learned.

Repair. Fidelity is higher early than late, so an earlier state is the more trustworthy one. We take the model’s belief k steps back, roll it forward under the moves that actually occurred, and inject the difference:

$$z_{\text{rep}} = T(\cdots T(z_{t-k}, x_{t-k+1}) \cdots, x_t), \quad h_t \leftarrow h_t + \eta W(z_{\text{rep}} - z_t).$$

No step reads the board.

3 Results

3.1 Joint training degrades the model

Trained jointly with the language model, the consistency objective raises validation loss from 3.0287 for the baseline to 3.1538 at weight 0.1 and 3.4250 at weight 0.5. The consistency task itself is learned equally well at both weights, 0.975, so the larger weight buys damage and nothing else. This is the pattern reported for oracle-supervised state channels: the auxiliary objective improves the representation it targets while competing with the task. Everything below therefore uses a frozen model.

3.2 The operator learns its task

Fitted on frozen activations, the one-step operator reaches 0.975 accuracy and the multi-step operator 0.996. Neither sees the board at any point. By the measure the objective defines, both succeed.

3.3 The rollout is worse than doing nothing

To ask whether a rollout is a better state than the model’s current one, we fit a read-out from the state subspace to board occupancy and compare three states at the same position. The read-out is a measuring instrument only; it is fitted on a disjoint split and never influences what is injected, so the repair stays oracle-free.

Table 2: Test-time injection of the repaired state. Illegal top-1 move rate by depth, with no-op giving the base rate, stale injecting the earlier state without rolling it forward, and random a norm-matched direction. Nothing separates them.

Ply	no-op	repair	stale	random
20–30	0.0386	0.0386	0.0378	0.0409
30–40	0.0885	0.0894	0.0780	0.0853
40–50	0.1667	0.1693	0.1631	0.1675
50–60	0.2227	0.2217	0.2237	0.2267
60–80	0.2965	0.2950	0.2972	0.2957

With the one-step operator the rollout is far worse than the model’s current state, 0.408 against 0.499 in the opening, and worse even than the unrolled stale state at 0.427. Applied twelve times, an operator with high one-step accuracy compounds its errors until the board is destroyed. One-step consistency constrains a single application and says nothing about composition.

Training the operator on rollouts directly closes most of that gap. The multi-step rollout reaches 0.510 against 0.427 for the stale state, so it is genuinely carrying state forward rather than decaying to it. It still never exceeds the model’s own current state, 0.530 in the opening and 0.362 at depth.

3.4 Repair therefore does nothing

Injecting the repaired state changes the illegal-move rate by nothing distinguishable from no-op across 6,033 scored positions, and the stale and norm-matched random controls are equally inert. The injection is small relative to the residual stream, at 0.037 of its norm, but a larger one would not help: what is being written is worse than what is already there.

3.5 Why no such method can work

The result is not a tuning failure. A model at position t attends over the entire prefix, so every token that determined the current board is still visible to it. Its state at $t - k$ is a lossy function of a strict subset of those same tokens. Rolling that state forward reconstructs, at best, a degraded version of something the model could already compute, and the reconstruction is performed by a linear operator far weaker than the network that produced the original.

Context rot in this setting is therefore not a storage failure. The information has not fallen out of the window; it is present and attendable throughout. What degrades is the computation performed over it. That distinction has a practical consequence: interventions that re-present or re-encode information the model can already see should not be expected to help, and the useful direction is to change what computation is available at depth rather than what information is.

3.6 Detection succeeds where repair fails

The same machinery yields something usable. If the representation has drifted, the model’s current state and its rolled-forward earlier state disagree, and that disagreement is computable at inference from activations and tokens alone. No oracle is used at any stage, including for training the operator, which distinguishes it from probe-based monitors that need supervised state labels to fit.

Over 60,056 positions from 1,044 games, consistency error predicts an illegal next move at AUC 0.5795 [0.5713, 0.5872], below the model’s own confidence at 0.6329 [0.6270, 0.6390]. It is the weaker signal alone and it is not redundant: fitted on training games and evaluated on held-out

Table 3: Oracle-free drift detection. Illegal-move rate among positions retained at each coverage, on held-out games, with the combination fitted on training games only. Consistency error is the weaker signal alone and is not redundant with the model’s own confidence.

Coverage	model confidence	consistency error	combined
100%	0.1989	0.1989	0.1989
90%	0.1825	0.1886	0.1841
80%	0.1741	0.1788	0.1723
70%	0.1631	0.1740	0.1582
60%	0.1543	0.1665	0.1447
50%	0.1423	0.1604	0.1283

games, the two together reach 0.6482 against 0.6299 for confidence alone, with standardised weights of +0.585 and +0.212. Used to abstain, the combination lowers the illegal rate among retained positions from 0.1423 to 0.1283 at half coverage, from a base rate of 0.193.

The gain is modest. Its interest is that it costs no oracle at any point, so it transfers to settings where no rules engine exists to supervise a probe.

4 Limitations

- L1** One model family, one domain, and a parameter range far below any deployed system. The mechanism argument in Section 2 is general, but the measurements are not.
- L2** The negative result concerns repair from the model’s *own* representations. It does not rule out repair from an external memory, from a re-encoded context, or from a second model.
- L3** The transition operator is linear. A stronger operator might roll forward better, though the information argument suggests the ceiling is the model’s own current state rather than something above it.
- L4** The read-out used to evaluate rollout quality is supervised, and is a measuring instrument only. The repair and detection procedures themselves use no oracle.
- L5** Detection is evaluated on one model and one operator; the interval is clustered on games but not across seeds.

5 Conclusion

Repairing a decayed state from a model’s own earlier representation does not work, and the reason is that there is nothing there to recover. The tokens remain in view; the earlier state is a lossy summary of them; and a small operator rolling that summary forward cannot outperform the network’s own current computation over the full prefix. Context rot, at least here, is a failure to compute rather than a failure to store.

That leaves a narrower but usable result. The mismatch between what the model currently believes and what its earlier belief implies is measurable at inference without any oracle, and it tells you something about when to distrust the next action that the model’s own confidence does not.

References